Classic 3 W / 1.2 W: Sprint calibrated; independent Fox 10 mW
NORMAL: protected 4S, 12.0–16.8 V. FOX–only: controller 3.3 V from 2S.
No 2 m changes. No production PCB. No measured prototype results.
Inter–sheet connections use explicit named global nets.
Read docs/M3_80M_SCHEMATIC_REVIEW.md before use.

01 Controller interface and local source

File: 01_source_interface.kicad_sch

02 Autonomous lease, inhibit and rearm

File: 02_interlocks.kicad_sch

03 Permission logic and latched power modes

File: 03_safety_logic.kicad_sch

04 Rearm latch, selected rails and key permits

File: 04_arm_key.kicad_sch

05 4S input, one buck and independent rail isolation

File: 05_power.kicad_sch

06 Reference, pack voltage and thermal/current trips

File: 06_analog_protection.kicad_sch

06b Independent threshold dividers

File: 06b_thresholds.kicad_sch

07 PA current trip and case–temperature interlock

File: 07_current_thermal.kicad_sch

08 Proven normal RF core and QCX–derived drain shaping

File: 08_normal_rf.kicad_sch

08b Shared key–tail and disarm reset

File: 08b_key_tail.kicad_sch

09 Original 80 m elliptic filter and normal port

File: 09_normal_filter.kicad_sch

10 Independent Foxoring shaper and rail–following clock buffer

File: 10_fox_rf.kicad_sch

11 Separate Foxoring filter, calibration pad and protected compact port

File: 11_fox_filter.kicad_sch

ARDF–TX / review only

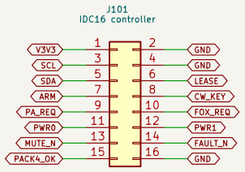
Sheet: /
File: RF80.kicad_sch

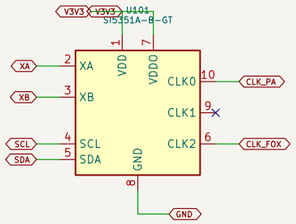
Title: ARDF–TX 80 m RF module

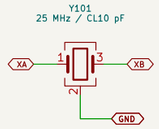
Size: A3	Date: 2026–10–14	**Rev: M3A–R0**
KiCad E.D.A. 10.0.6		Id: 1/14

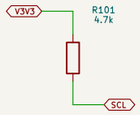
01 Controller interface and local source

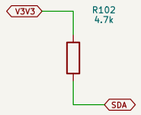
One local Si5351; no MCU/RTC/UI on RF PCB. Shared 3.3 V stays powered during inhibited I2C setup.
Controller wide-input regulator remains a proposal: evaluation fixture supplies bounded 3.3 V separately.

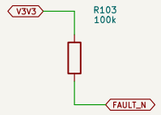


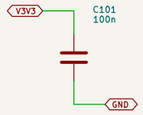


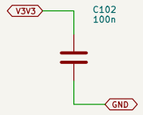


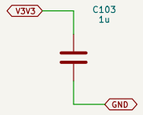


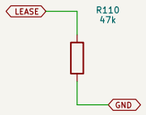


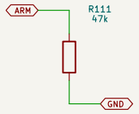


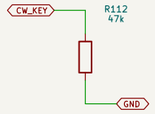


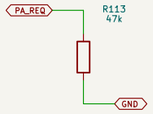


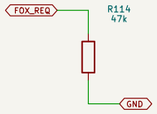


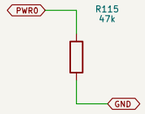


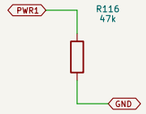


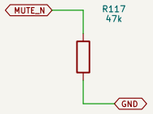


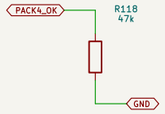


























Logical contract: 3.3 V only; V3V3 3.135–3.465 V, absolute connector maximum 3.6 V.
RoutePAcurrentthroughdedicatedJ301return; checkcontrollerground–harnesscurrent. PA raw power has separate J301 and return.
LEASE 20 Hz software-progress toggles; ARM low before changing frequency, path or power.
Any SCL activity while armed trips the local latch. FAULT_N open drain, low when unarmed.
Do not connect TTL 5 V inputs or an unqualified controller raw supply to this connector.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization
ARDF-TX

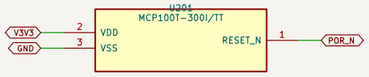
Sheet: /01 Controller interface and local source/
File: 01_source_interface.kicad_sch

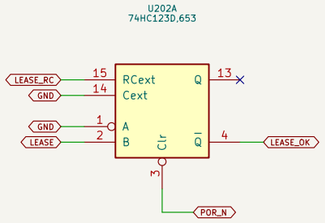
Title: 01 Controller interface and local source

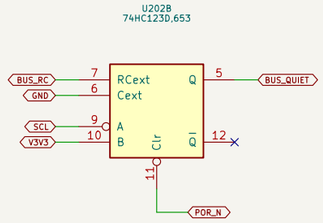
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 2/14

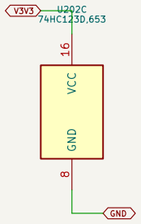
02 Autonomous lease, inhibit and rearm

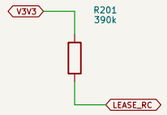
Healthy-only asynchronous reset. Latched fault cannot recover while ARM remains high.
ARM RC/Schmitt delay releases CLR before clocking. Requests exclusive; both asserted trips.

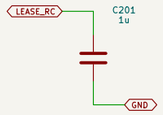


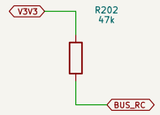


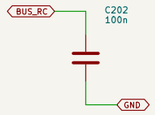


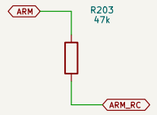


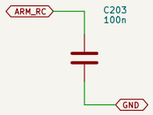


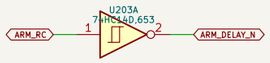












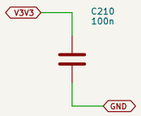


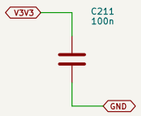


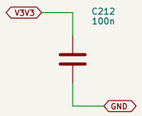


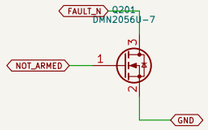


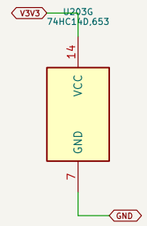












U202 first half: positive LEASE edge, /A held low. Second half: falling SCL with B high.
No blind ISR heartbeat: controller advances LEASE only after schedule and bus-state checks.
Local mute is independent of firmware: hardware does not verify the requested frequency value.
Normal key-up leaves driver active through the measured drain-envelope tail. ARM/MUTE kill immediately.





M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization
ARDF-TX

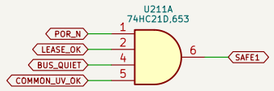
Sheet: /02 Autonomous lease, inhibit and rearm/
File: 02_interlocks.kicad_sch

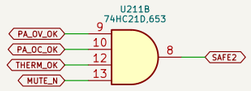
Title: 02 Autonomous lease, inhibit and rearm

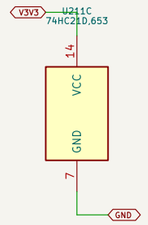
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 3/14

03 Permission logic and latched power modes

NOT a second contest processor. Hard gates combine measured electrical health and controller requests.





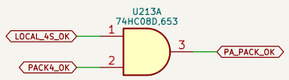


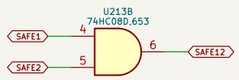


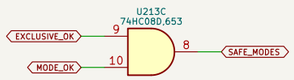


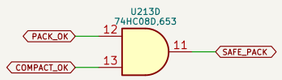


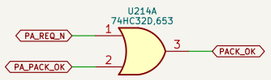




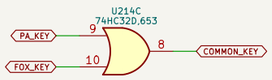


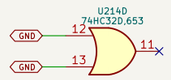


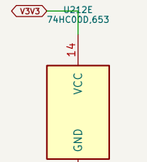


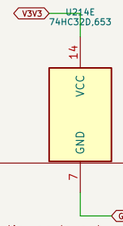


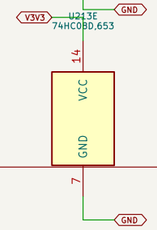


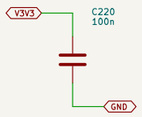


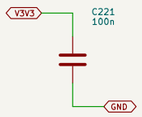


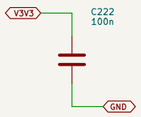


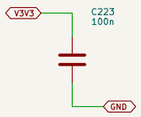












M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

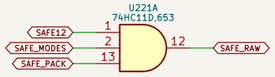
ARDF-TX

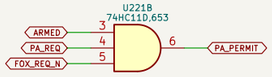
Sheet: /03 Permission logic and latched power modes/
File: 03_safety_logic.kicad_sch

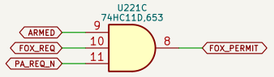
Title: 03 Permission logic and latched power modes

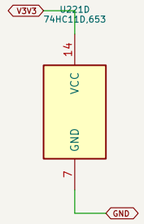
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 4/14

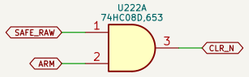
04 Rearm latch, selected rails and key permits

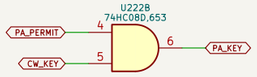


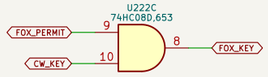


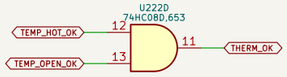


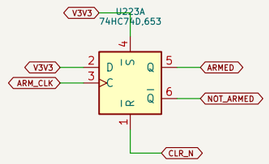


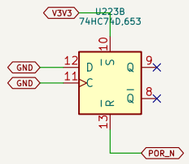


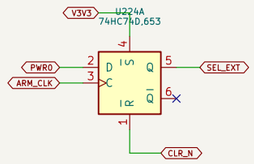


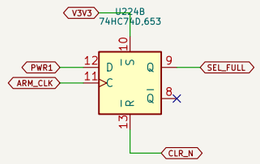


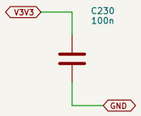


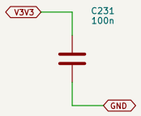


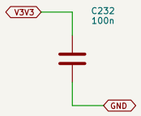


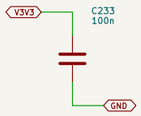


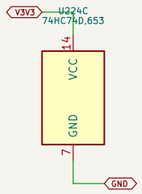


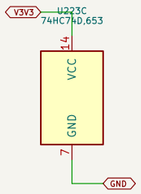


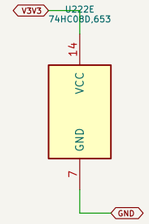












PWR1/PWR0: 00 Sprint (start4.68V); 01 extended (start5.58V); 10 full (start7.96V); 11 fault.
Modes latch on delayed ARM edge and reset to Sprint on fault. Changing PWR while armed does not change rail;
changing to11 still faults. Retuning/I2C, stopped heartbeat, MUTE low, UV/OV/OC/hot/open NTC clear ARMED.
Rearm: ARM low >=20ms, health stable + I2C quiet, requests/mode fixed; ARM high then wait >=250ms beforefirstKEY.
No auto-rearm after fault with ARM stuck high. Verify asynchronous recovery/Schmitt delay with real parts.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

ARDF-TX

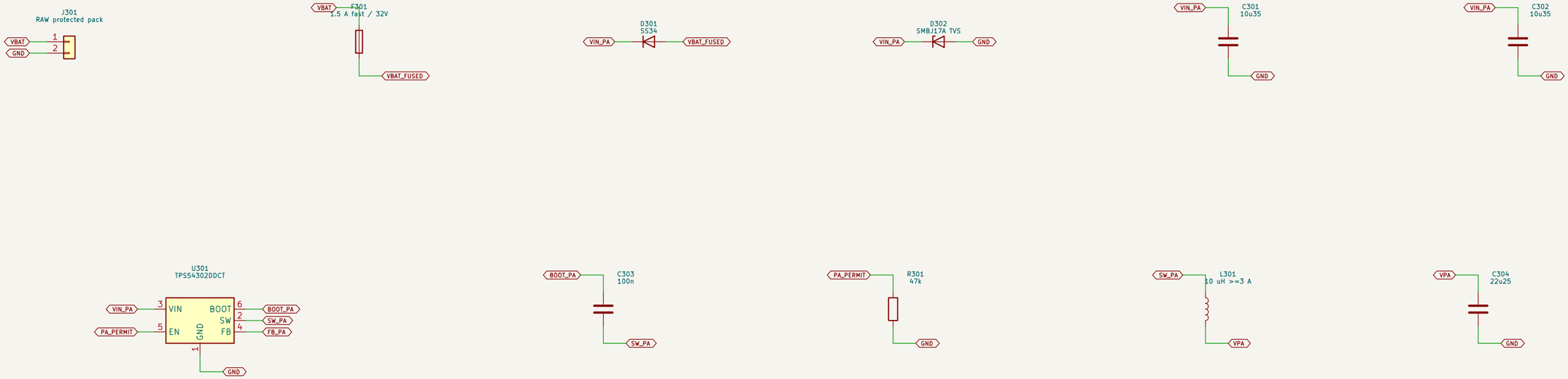
Sheet: /04 Rearm latch, selected rails and key permits/
File: 04_arm_key.kicad_sch

Title: 04 Rearm latch, selected rails and key permits

Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 5/14

05 4S input, one buck and independent rail isolation

Shared J301/F301/D301 input barrier; converter/loadswitch NORMAL only. Protected4S:12.0–16.8V.
Fox-only retains input barrier/sense bypass, omits buck/loadswitch and watt-rail parts;2S powerscontroller+input sense.



M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization
ARDF-TX

Sheet: /05 4S input, one buck and independent rail isolation/
File: 05_power.kicad_sch

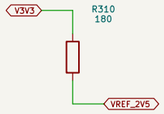
Title: 05 4S input, one buck and independent rail isolation

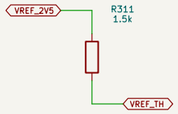
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 6/14

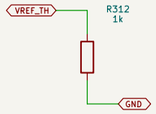
06 Reference, pack voltage and thermal/current trips

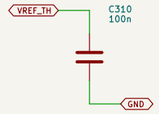
Real electrical trips; no RF power detector or continuous telemetry. Comparator inputs stay below1.5V.
Shared reverse-protected J301 pack input retained for Fox-only; all PA converter/driver/final components DNP.

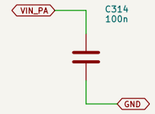


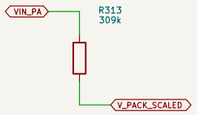


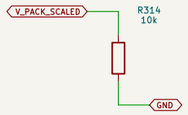


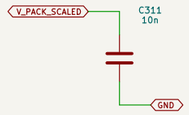


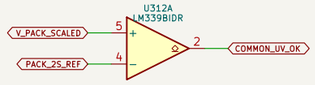


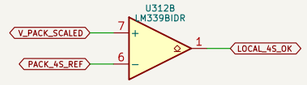


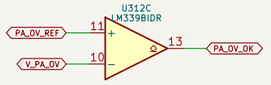


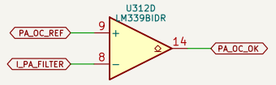


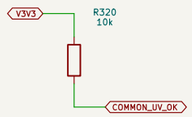


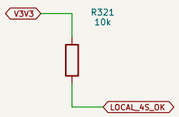


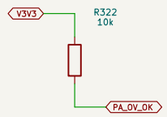


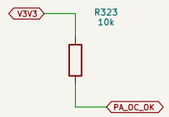


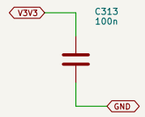


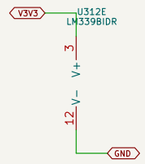


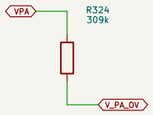


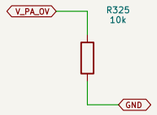












Nominal backup: commonUV5.810V/4Sbackup11.070V postreversebarrier; railOV8.973V.
Controller operational endpoint remains6.0/12.0V with hysteresis; cell-group BMS external.
Local pack sensing does not prove weakest-cell voltage. Hysteresis implemented by fault latch;
rearm only after input exceeds qualified margin. Fox-only: VPA node is grounded via R325/R324, not floating.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

ARDF-TX

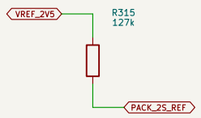
Sheet: /06 Reference, pack voltage and thermal/current trips/
File: 06_analog_protection.kicad_sch

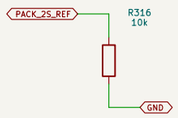
Title: 06 Reference, pack voltage and thermal/current trips

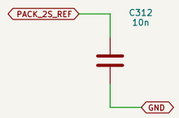
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 7/14

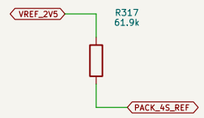
06b Independent threshold dividers

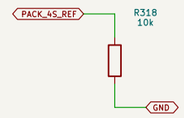
Independent reference dividers from regulated TL431 node, not a loaded shared half-reference.
Pack and PA voltage signals divideby31.9; INA current signal divides by4. No comparator pin exceeds1.05V
for28V input/rail or3.465V current-sense saturation. Thermal divider ratios sharetheir own<1.0V node.

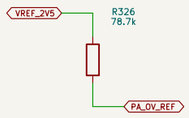


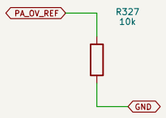


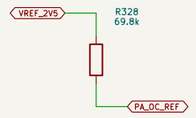


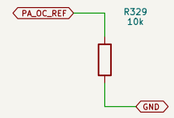












Calculated at2.495V reference: commonUV5.810V protectedinput;4Sbackup11.070V protectedinput;
OV8.973V rail; currenttrip1.251A. Addmeasured5S34drop forrawpackthreshold.
Controller operation stillstopsatapproximately6.0/12.0V; cell-group BMSexternal.
Referencebias180ohm supports>=1mA TL431cathodecurrentafterallparallelreferencebranchloads.
Cold/hotreference, comparatoroffset/inputbias, resistor tolerances andtripresponse mustbe measured.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

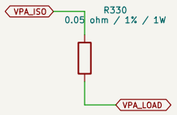
Review schematic; no layout or ordering authorization
ARDF-TX

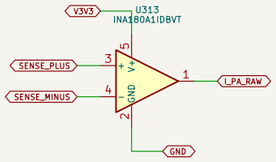
Sheet: /06b Independent threshold dividers/
File: 06b_thresholds.kicad_sch

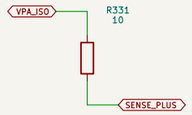
Title: 06b Independent threshold dividers

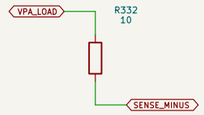
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 8/14

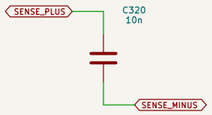
07 PA current trip and case-temperature interlock

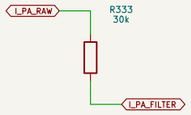


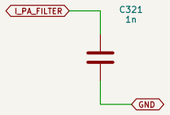


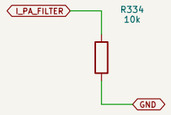


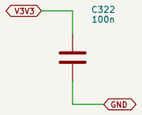


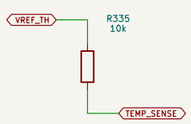


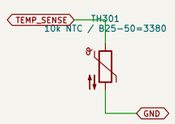


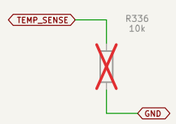


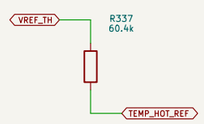


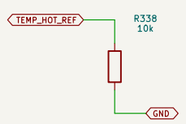


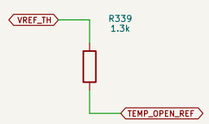


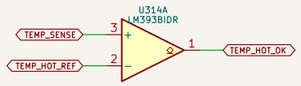


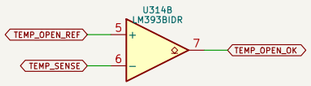


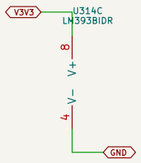


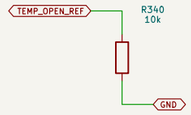


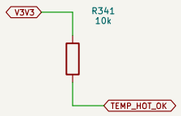


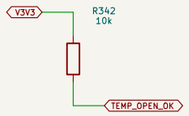


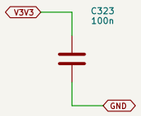














Current trip nominal1.251A (20*0.05ohm=1V/A); targetfaultmute<=100us; dividergain0.25V/A:7.5ustau.
Temperature window detects hot/short and detached/open sensor; nominalhot ~80C case.
Values/thermal contact, comparator tolerance, filtering and trip latency are qualification gates.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

ARDF-TX

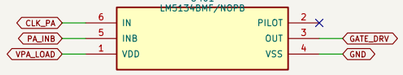
Sheet: /07 PA current trip and case-temperature interlock/
File: 07_current_thermal.kicad_sch

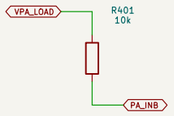
Title: 07 PA current trip and case-temperature interlock

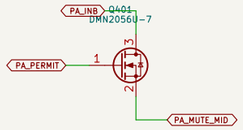
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 9/14

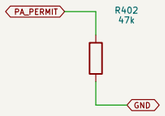
08 Proven normal RF core and QCX-derived drain shaping

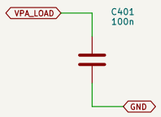
RF core adapted from OpenARDF SignalSlinger HW3.5 (MIT); shaper principle from QCX/W6JL.
Keep clock running during graceful key tail; INB immediately mutes on fault. This is NOT the VHF AM modulator.

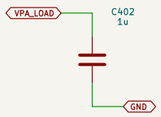


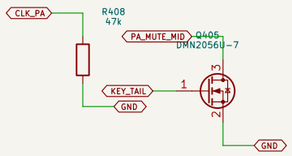


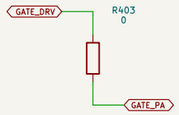


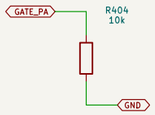




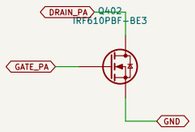


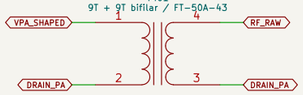


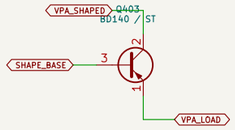


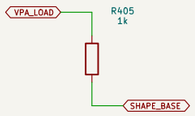


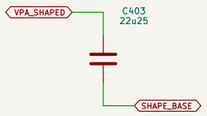
No PA/drain reservoir after shaper. Driver reservoir is upstream.
9+9 series autotransformer: effective voltage step2:1, load-12.5ohm beforefilter.
Transformer drawing is coupled coils; center join is external and explicit.
All components on this sheet DNP on Fox-only. No watt-to-Fox RF switching.

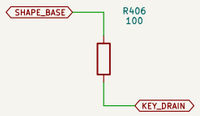


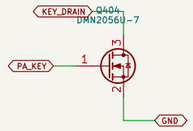


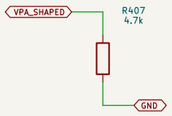




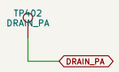


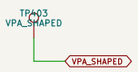


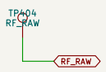












M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization
ARDF-TX

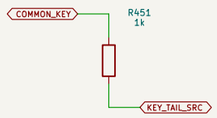
Sheet: /08 Proven normal RF core and QCX-derived drain shaping/
File: 08_normal_rf.kicad_sch

Title: 08 Proven normal RF core and QCX-derived drain shaping

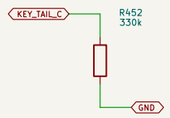
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 10/14

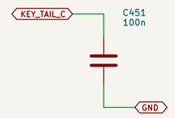
08b Shared key-tail and disarm reset

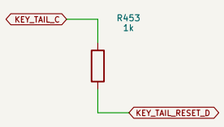
Path permission alone must not startRF during drain-integrator charging.
Normal series INB sink needsPA_PERMIT andKEY_TAIL; FoxRFmute also requiresFOX_PERMIT andKEY_TAIL.
COMMON_KEY is OR ofpermittedPA_KEY/FOX_KEY. Exclusivepathlogic prevents simultaneous RF.

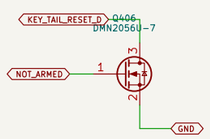
















RCtailtau33ms; actualholddependsdiodedrop/HC14hysteresis. Benchqualify10-80mshold,
longerthanthe3-8msRFdecay. ShorterfallcanclipRF; excessiveholdraisesidle/leakage.
ARMlow>=20ms clearsmemory. Afterarming wait>=250msbeforefirstkey, thenverifysettleddrain.
A fault bypassesgracefuldecay: PApermission/driver/railoff immediately.
RFcalculation usesgatechargefortheentire20%scheduledslot, notonlyCWmarktime.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization
ARDF-TX

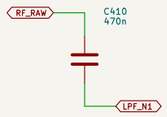
Sheet: /08b Shared key-tail and disarm reset/
File: 08b_key_tail.kicad_sch

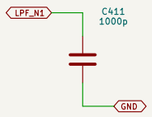
Title: 08b Shared key-tail and disarm reset

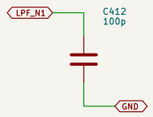
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 11/14

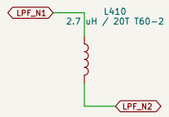
09 Original 80 m elliptic filter and normal port

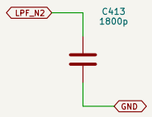
Retain SignalSlinger 80m values/topology: series470nF, two2.7uH, 1100/1800/1100pF,91pF acrosssecondcoil.
Split1100pF into1000+100pF for obtainable capacitors. Coil recipe is mandatory; no generic power inductors.

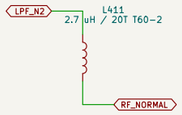


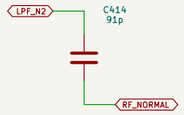


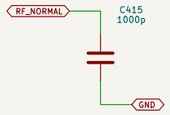


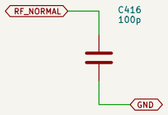


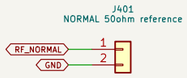


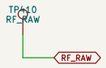


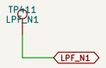


















Reference50ohm power: J401, calibrated adapter+dummyload. No antenna gain/ERP conversion on80m.
Long antenna matcher remains separate: measured verticalwire+defined counterpoise, L-match/variometer.
Normal match START range15–50uH fromDF1FO: exacttap/cap values requireinstalledimpedance.
A screwterminal can save~4EUR versus Amphenol132134 SMA; reference-plane calibration remains mandatory.
Harmonics/spurs must be measured after this filter across3.510–3.600MHz and everycalibratedsetting.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

ARDF-TX

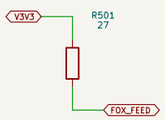
Sheet: /09 Original 80 m elliptic filter and normal port/
File: 09_normal_filter.kicad_sch

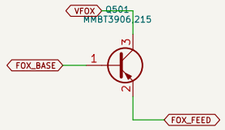
Title: 09 Original 80 m elliptic filter and normal port

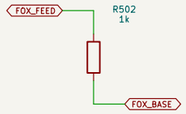
Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 12/14

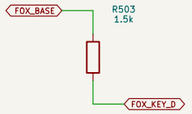
10 Independent Foxoring shaper and rail-following clock buffer

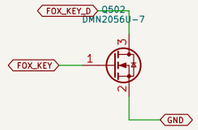
TinyFox three-gate mechanism with stronger SN74AC00DR; new rail-following NPN input + QCX-style low-current envelope.
No electrical RF connection to RF_RAW/RF_NORMAL. Nominal10mW is calibrated at separate J502, not inferred.

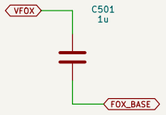


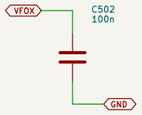


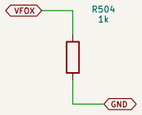


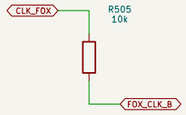


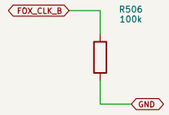


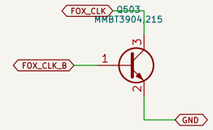


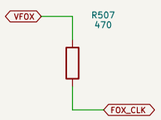




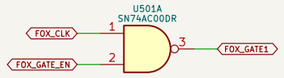


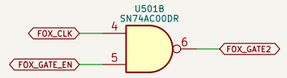






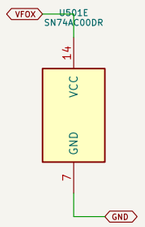




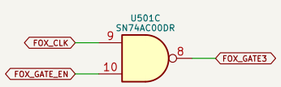


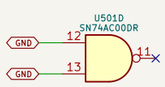


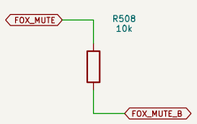


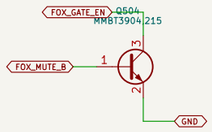


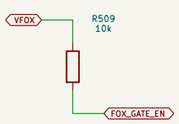












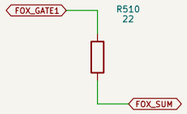
M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

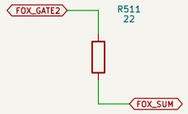
Review schematic; no layout or ordering authorization
ARDF-TX
Sheet: /10 Independent Foxoring shaper and rail-following clock buffer/
File: 10_fox_rf.kicad_sch

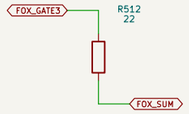
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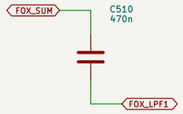
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KiCad E.D.A. 10.0.6		Id: 13/14

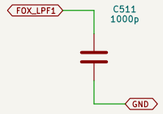
11 Separate Foxoring filter, calibration pad and protected compact port

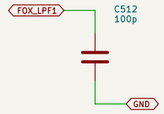


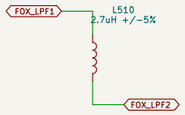


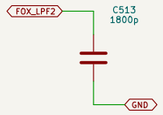


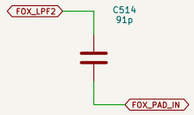


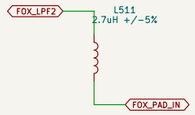


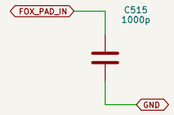


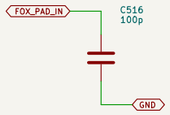


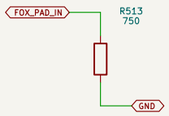


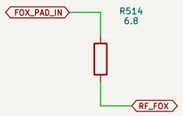


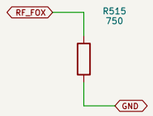


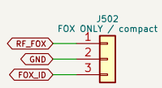


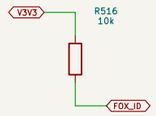


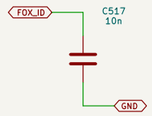




















J502 reference plane: pin1 RF/pin2 definedreturn; use shorttwistedpair coaxadapter sharedfixture.
Compactaccessory: -30 cmradiator+definedcompactreturn; tune independentloading/couplingcartridge.
StartingTinyFoxreference100uH plus1.6–15pF: effectiveC5–15pF mayrequire134–402uH.
Exact30cm geometry/tap/coil/capvalues OPEN untilantenna impedance measurement; no Classicwire substitute.
Nominal10mW in50ohm and30maudible/250minaudible are TWO distinct acceptance tests.

M3A-R0: REVIEW / ALL hardware measurements NOT RUN / no layout or procurement.

Review schematic; no layout or ordering authorization

ARDF-TX

Sheet: /11 Separate Foxoring filter, calibration pad and protected compact port/
File: 11_fox_filter.kicad_sch

Title: 11 Separate Foxoring filter, calibration pad and protected compact por

Size: A3	Date: 2026-10-14	**Rev: M3A-R0**
KiCad E.D.A. 10.0.6		Id: 14/14